Yehan Zhang

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RESEARCH INTERESTS

LLM Agents, Trustworthy AI, Lifelong Learning, Multimodal Understanding

EDUCATION

South China University of Technology, Guangzhou, China — Sept 2023 — Jul 2027
Bachelor of Science: Data Science and Big Data Technology — GPA: 3.67/4.00, TOEFL: 5.0/6.0
Selected Coursework: Machine Learning (4.00/4.00), Deep Learning for Computer Vision (4.00/4.00), Discrete Mathematics (4.00/4.00), 3D Vision Intelligence (4.00/4.00), Calculus (4.00/4.00).

PUBLICATIONS

1. **Yehan Zhang**, Huaidong Zhang, Xuandi Luo, Shengfeng He. — Sole First Author
   Healthcare Insurance Fraud Detection via Continual Fiedler Vector Graph Model.
   The Fourteenth International Conference on Learning Representations (ICLR 2026).
   Accepted as Poster.

2. **Yehan Zhang**[*], Qianqi Yan[*], Hang Yin, Ching-Chen Kuo, Shan Jiang, Ming Ouyang, Yijie Liu, Xin Eric Wang. — Co-first Author
   AgentDecept: Benchmarking Deceptive Behavior in Tool-Using LLM Agents under Pressure.
   Under Review.

3. **Yehan Zhang**, Hongbo Xu, Zhou Liu, Yongwei Nie, Fei Ma. — Sole First Author
   Coverage-Aware Coarse-to-Fine Retrieval for Multimodal Long-Document QA.
   Preprint.

4. Xuandi Luo, Huaidong Zhang, **Yehan Zhang**, Shengfeng He. — Third Author
   DynLearn: Steering Task-Specific Learning Dynamics in Exemplar-Free Class-Incremental Learning.
   Under Review.

RESEARCH EXPERIENCE

Trustworthy AI in Agents Behavior — May 2026 — Sept 2026
Research Intern — *University of California, Santa Barbara*

- Built an evaluation framework to distinguish deceptive behaviors in tool-using LLM agents under pressure.
- Designed six pressure categories and paired pressure controls, constructing an evaluation set of 477 pressure tasks.
- Combined tool-execution evidence auditing with linear probes to identify deceptive behavior.

Multimodal Understanding in Long Documents — Nov 2025 — Mar 2026
Research Intern — *Guangdong Laboratory of Artificial Intelligence and Digital Economy*

- Developed CERAG to retrieve complete evidence for multimodal long-document question answering.
- Built a multimodal retrieval and reranking pipeline with coverage-aware loss to retain key evidence.
- Achieved state-of-the-art performance on the MMLongBench-Doc and other three document understanding benchmarks.

Lifelong Learning in Fraud Detection — May 2025 — Sept 2025
Research Intern — *South China University of Technology*

- Propose a model for fraud detection under label scarce and non-stationary conditions.
- Develop a FiedlerVector-guided graph autoencoder that leverages global topological signals.
- Introduce a Subgraph Attention Fusion module for unsupervised online learning, enabling dynamic risk representation and continual adaptation to evolving fraud patterns.

SKILLS

- **Programming:** Python, Codex.
- **Language:** English, Chinese, Spanish.